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UFO

Atomized Chain-of-Evaluation Achieves 15% Better Human-Correlation for Multi-Modal Image Generation by Evaluating All Conditions Simultaneously

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Multi-modal image generation is evaluated by checking each condition independently—a fundamentally wrong strategy when conditions interact and conflict. *UFO* introduces Atomized Chain-of-Evaluation: decompose each condition into fine-grained Atomic Evaluation Units, order them by dependency, call purpose-built verifiers, and aggregate. On the new UFO-Bench (660 cases, 7 categories, 3 difficulty tiers), *UFO* achieves Spearman $\rho=0.689$ with human rankings—a 15.25% improvement over the best prior metric.

AT A GLANCE

Problem: Existing multi-modal generation metrics evaluate each condition (text, reference image, layout, style, identity) in isolation, ignoring interactions that determine human preference.

Why it matters: Subject-driven customisation and layout- conditioned generation are competitive research areas; unreliable metrics stall progress and misrank models.

Method: Decompose each condition into Atomic Evaluation Units (AEUs); categorise by modality-relevance class; order into a chain; call structured (detector-based) or generic (MLLM) verifiers per AEU.

Result: Spearman $\rho = 0.689$ vs. $\rho = 0.598$ (VIEScore); +15.25% average improvement. Largest gain on hard (conflicting-condition) tier.

Evidence: UFO-Bench: 660 human-annotated cases, 5 raters each, Krippendorff $\alpha=0.73$; evaluated on CLIP-T, DINO-I, HPS-v2, VIEScore, and six other baselines.

The Big Picture

Subject-driven text-to-image models accept a reference image (the subject), a text prompt, a layout specification, and optionally a style or identity condition—all simultaneously. The output is evaluated for how well it aligns with all provided conditions at once.

Current metrics fail this task. CLIP-T measures text-image cosine similarity; DINO-I measures instance identity similarity; they are then averaged into a scalar. But this average hides condition priority: if a prompt says “make the dog hold a red umbrella” and the model produces a dog without an umbrella, no averaging of text and identity scores captures this failure correctly. Humans, presented with the same output, immediately rank it below a generation that at least includes the umbrella.

UFO treats evaluation as a structured reasoning problem: enumerate what must be verified, verify each part with the best available tool, and combine respecting priority and dependency.

Why Existing Approaches Fall Short

Embedding-based metrics (CLIP-T, DINO-I, CLIP-I) measure condition-specific cosine similarity in a shared embedding space. Their scores are invariant to fine-grained compositional failures; a tiger wearing a hat and a tiger without a hat may have nearly identical CLIP scores with the same text prompt.

MLLM-based metrics (VIEScore, Pick-Score) use a multimodal language model as a judge but ask it to evaluate all conditions holistically in a single prompt. The model must internally reason about interactions it was not specifically trained to handle; its judgments are inconsistent on conflicting-condition cases.

Human Preference Score (HPS-v2) is trained on general human preferences that reflect aesthetics and realism as much as condition alignment, making it poorly calibrated for the specific task of measuring omni-condition satisfaction.

Core Mechanism

AEU decomposition. For a set of conditions $C = \{c_1, \dots, c_K\}$ (text prompt c_T , reference image c_I , layout c_L , style c_S , ...), each condition is parsed into binary claims about one attribute:

$$\{u_{k,j}\} = \text{Parse}(c_k), \quad u_{k,j} : \text{“attribute } a_{k,j} \text{ of } c_k \text{ satisfied?”}$$

Chain ordering. AEUs are sorted topologically over a dependency graph G_C where edges encode precedence (subject identity must be located before subject pose can be verified). Mandatory conditions come first; preferred and contextual follow.

Specialised verifiers.

- *Text attributes:* MLLM with targeted yes/no prompts.
- *Identity conditions:* ArcFace (persons) or DINOv2 cosine similarity (objects).
- *Geometric conditions:* DWPose (body pose), Grounding-DINO + SAM2 (object placement),

DepthAnything (depth layout).

- *Style conditions*: CLIP style embedding vs. reference.

Weighted aggregation. Mandatory AEU's have weight 1.0, preferred 0.7, contextual 0.3:

$$\text{OCA}_{\text{UFO}}(x, C) = \sum_{k,j} r_{k,j} \cdot V_{k,j}(x, c_k, a_{k,j}).$$

Technical Formulation

Let $\mathcal{U} = [u_1, \dots, u_n]$ be the ordered chain of AEU's and V_i the verifier for AEU u_i . The OCA score is:

$$\text{OCA}(x, C) = \frac{\sum_{i=1}^n r_i V_i(x)}{\sum_{i=1}^n r_i}, \quad r_i \in \{1.0, 0.7, 0.3\}.$$

Spearman correlation with human rankings:

$$\rho = 1 - \frac{6 \sum_i d_i^2}{n(n^2 - 1)},$$

where d_i is the rank difference between UFO scores and human preference rankings over the $n = 660$ UFO-Bench cases.

For text prompt c_T with P phrases, each phrase p is parsed into an AEU with attribute type (object, property, spatial, action, count) and relevance class. The dependency graph adds edges from objects to their attributes (the object must be localised before its colour or pose can be verified).

Learning or Inference Procedure

UFO is training-free. Inference steps:

1. Parse all conditions C into AEU's; extract attribute types and relevance classes.
2. Build dependency graph G_C ; topologically sort into chain $\mathcal{U}$.
3. For each AEU u_i : call verifier V_i with the generated image x and condition c_k .
4. Compute weighted aggregation $\text{OCA}_{\text{UFO}}(x, C)$.

New condition types are added by registering an AEU parser and verifier function—no retraining. Total evaluation time per image is approximately 4 seconds on a single GPU.

What the Guarantee Says

UFO provides a theoretical lemma: under conditional independence between conditions, the chain-of-evaluation score is an unbiased estimator of the true omni-condition alignment. For the general (dependent) case, the mandatory-first ordering minimises the expected rank correlation error by prioritising the conditions humans weight most heavily, which the paper verifies empirically across annotation tiers.

Experimental Findings

Rank correlation with human preferences on UFO-Bench (660 test cases, 5 annotators each):

Metric	Spearman ρ	Pearson r	Kendall τ
CLIP-T	0.421	0.398	0.312
DINO-I	0.389	0.374	0.288
HPS-v2	0.521	0.508	0.401
VIEScore	0.598	0.581	0.463
UFO	0.689	0.672	0.537

By difficulty tier: Easy $\rho = 0.741$; Medium $\rho = 0.693$; Hard $\rho = 0.701$. UFO's advantage is largest on hard cases, where conflicting conditions are the norm and independent metrics collapse.

Ablations and Interpretation

MLLM for all AEU's (no structured verifiers): ρ drops to 0.631; geometric and identity AEU's are specifically hurt, confirming that dedicated detectors are needed for these types.

Random AEU ordering: $\rho = 0.657$; mandatory-first ordering resolves conflicting conditions more consistently.

Uniform weights ($r_i=1$ for all): $\rho = 0.672$; graded weights align with human priority on medium and hard tiers.

Sentence-level vs. phrase-level AEU's: Phrase-level AEU's improve ρ by 0.042; sub-phrase (word-level) adds only 0.003 at $2\times$ compute.

Model comparison on UFO-Bench: FLUX.1 outranks SDXL by 0.08 UFO score on hard cases but only 0.02 by CLIP-T average, illustrating UFO's superior discriminative power for state-of-the-art models that are close to each other on simple metrics.

Reference

Danning Zhang, Yijing Lin, Shuhan Zhuang, Mengqi Huang, Shaojin Wu, Shancheng Fang, Zhendong Mao. **UFO: Chain-of-Evaluation for Omni-Condition Alignment in Multi-Modal Image Generation.** arXiv:2609.12397, September 2026. ICML 2026. <https://arxiv.org/abs/2609.12397>